

GIS-Based Feasibility Mapping and Energy Demand Analysis for Microreactor Deployment in the Philippines

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ABSTRACT

The Philippines operates 132 Small Isolated Island Grids (SIIGs) supplying over 1.2 million households in off-grid communities through costly, price-volatile diesel generation. In February 2026, the Universal Charge for Missionary Electrification (UC-ME) registered its highest rate increase in five years – rising 38% year-on-year – as global diesel prices surged following the closure of the Strait of Hormuz (Climate Smart Ventures, 2026). Microreactors (1–20 MWe) – factory-built, barge-transportable, with 8–20 year refueling cycles – offer a baseload alternative that eliminates diesel logistics entirely (Idaho National Laboratory, 2025). This study presents the first machine learning-augmented microreactor siting framework for the Philippine archipelago, integrating GIS hazard exclusion mapping, multi-criteria suitability scoring, Random Forest (RF) classification, K-Means clustering, and Energy Coverage Ratio (ECR) analysis across 11 open-access datasets. 27.8% of Philippine land area is excluded by hazard constraints; 200 candidate sites are identified nationwide; the RF classifier achieves 87.8% accuracy (5-fold CV: $89.0\% \pm 1.8\%$) with zero false negatives for suitable sites; SHAP analysis confirms distance to population centers as the dominant siting driver; and three deployment archetypes are identified (Coastal Island Hubs, Small Coastal Communities, Peri-Urban Grid Anchors). All top 20 ranked sites achieve a Balanced ECR (1.0–1.5), confirming that standard 1–2 MWe units are appropriate for the majority of candidate sites. The framework provides actionable GIS evidence aligned with the DOE Philippines' 2026 nuclear licensing roadmap and NPC-SPUG's Missionary Electrification Development Plan.

1. INTRODUCTION

1.1 The Philippine Off-Grid Energy Crisis – Current Situation (2025–2026)

The Philippine archipelago – 7,641 islands across 300,000 km² of land area – presents a structural electricity access challenge that conventional grid extension cannot economically solve. Three major interconnected grids (Luzon, Visayas, Mindanao) serve the main island groups, but 132 Small Isolated Island Grids (SIIGs) remain dependent on diesel-powered generators operated by the National Power Corporation's Small Power Utilities Group (NPC-SPUG) (Longdom, 2022). As of 2026, these SIIGs affect over 1.2 million households in the Philippines' most geographically isolated communities (Climate Smart Ventures, 2026).

The economic and energy security situation deteriorated sharply in early 2026. On March 2, 2026, the Strait of Hormuz – which handles roughly 20% of global fuel shipments from the Middle East – was closed due to geopolitical conflict, forcing tankers to take longer routes and causing the Mean of Platts Singapore (MOPS) gasoline price index to surge 13% in just three days (from USD 79.63/barrel on February 27 to USD 90.32 by March 2) (Climate Smart Ventures, 2026; Rappler, 2026). This price shock directly threatened the power supply of over 1.2 million households in Philippine off-grid communities, where 89.3% of installed off-grid generating capacity runs on oil (Rappler, 2026). End-user tariffs in SIIG areas already range from PHP 16–25/kWh – more than double the main grid rate of approximately PHP 10/kWh (Yale Clean Energy Forum, 2026).

The UC-ME subsidy mechanism that partially offsets these costs has itself become unsustainable. From 2022 to 2026, the UC-ME rate charged to all on-grid consumers grew at a

15% cumulative annual growth rate. In February 2026 alone, it registered its single largest increase in five years – rising 38% year-on-year – to allow NPC to settle arrears from fuel suppliers and prevent blackouts in SPUG areas (Eco-Business, 2026). The NPC is now undertaking its Accelerated Hybridization Program (AHP), which aims to launch 14 initial diesel-solar-battery hybrid plants with 25 additional sites by end of 2026, targeting at minimum a 20% reduction in diesel consumption and a full transition to renewables across all NPC-SPUG areas by 2030 (Eco-Business, 2026). The AHP represents a step forward but does not resolve the fundamental baseload reliability problem: solar and wind hybrid systems cannot guarantee 24/7 power supply without prohibitively expensive storage, particularly for islands where cloud cover and intermittency are chronic.

The study by Castro et al., (2024) modelling the transition of 208 Philippine off-grid island mini-grids to 100% renewable energy by 2050 found that an energy transition steered solely by market dynamics cannot achieve full renewable penetration – policy intervention and alternative baseload technologies are necessary. This gap – the need for a reliable, fuel-import-independent baseload technology for off-grid islands – is precisely the deployment context that microreactors are designed to fill.

1.2 The Philippines Nuclear Energy Trajectory – 2025–2026 Developments

In a landmark development for Philippine energy policy, President Bongbong Marcos signed Republic Act No. 12305, the Philippine National Nuclear Energy Safety Act, into law on September 18, 2025, formally establishing the Philippine Atomic Energy Regulatory Authority (PhilATOM) as an independent quasi-judicial regulatory body (PhilATOM Wikipedia, 2025). This legislation resolved the decades-long regulatory gap that had prevented nuclear power licensing in the Philippines. The DOE has stated that it plans to begin accepting nuclear power plant license applications in 2026, with PhilATOM overseeing the introduction of nuclear energy to the country's energy mix (Pinsent Masons, 2026).

The Philippines has identified 2026 as a 'turning point' for nuclear implementation, with a national target of 1.2 GW of nuclear capacity by 2032, rising to 2.4 GW by 2035 and 4.8 GW by 2050 (Pinsent Masons, 2026). The government has also completed a harmonised whole-of-government licensing and permitting flow chart for nuclear power plant projects, to be presented to prospective proponents alongside investment incentives (World Nuclear News, 2026a). In February 2026, the U.S. Trade and Development Agency (USTDA) committed USD 2.7 million in technical assistance for Meralco PowerGen Corporation (MGEN) to evaluate U.S. Small Modular Reactor (SMR) designs and develop an implementation roadmap (World Nuclear News, 2026b; Carbon Credits, 2026). At the regional level, ASEAN member states agreed in 2025 to integrate nuclear power into the ASEAN Plan of Action for Energy Cooperation (APAEC) 2026–2030, reflecting growing regional consensus that fossil-fuel-based energy cannot sustainably meet rapidly rising electricity demand (Pinsent Masons, 2026).

Critically for microreactor deployment specifically, Manila Electric Company (Meralco) – the Philippines' largest utility – signed a cooperation agreement with Ultra Safe Nuclear Corporation (USNC) in November 2023 to study the potential deployment of one or more microreactors (World Nuclear Association, 2026). Korea Hydro & Nuclear Power (KHNP) commenced a technical and economic feasibility study on potential rehabilitation of the Bataan Nuclear Power Plant in January 2025 (World Nuclear Association, 2026). These developments establish that microreactor deployment in the Philippines is no longer a theoretical proposition – it is an active policy and commercial consideration with institutional and financial backing.

1.3 Microreactor Technology for Off-Grid Islands

Microreactors are defined as nuclear reactors generating up to 20 MWe, with factory fabrication, passive safety systems, long refueling intervals of 8–20 years, and transportability by truck, rail, or barge (Idaho National Laboratory, 2025). These characteristics directly address the three core weaknesses of diesel generation in Philippine Island settings: recurring fuel logistics, price volatility exposure, and inability to guarantee 24/7 supply. A techno-economic evaluation established that microreactor electricity costs of USD 0.14–0.41/kWh overlap with diesel generation costs for remote and island communities, making microreactors cost-competitive at first-of-a-kind deployment (Mignacca & Locatelli, 2023). The IAEA has explicitly identified microreactors as suited for regions inaccessible to clean, reliable, affordable energy (IAEA, 2025).

1.4 Research Gap and Objectives

Despite the urgent policy context — a deteriorating diesel cost crisis, a new nuclear regulatory framework, active microreactor commercial interest, and ASEAN-wide nuclear policy momentum — no published study has systematically evaluated microreactor deployment feasibility in the Philippine archipelagic context using GIS-based spatial analysis. This study addresses that gap through two research questions:

1. Where are safe and suitable microreactor sites in the Philippines under real-world hazard, terrain, and infrastructure constraints?
2. Can candidate sites realistically meet local electricity demand under a demand-driven reactor sizing framework?

1.5. Scope and Limitations

- **DEM resolution:** 90m SRTM used for slope; may underestimate steep terrain in small islands. Analysis at 1 km resolution loses spatial detail for narrow coastal strips.
- **Demand estimation:** per-capita proxies from Pascasio et al. (2022), not metered surveys. Actual island consumption may differ.
- **MCDM weights:** scoring criteria weights and half-score distances are author-defined and not empirically calibrated. A sensitivity analysis was beyond project scope.
- **No economic analysis:** LCOE, NPV, capital cost, fuel cycle costs excluded. Technical suitability only.
- **No social acceptance:** community attitudes, BARMM indigenous considerations, and public trust — a key concern identified by the DOE (Nuclear Business Platform, 2025) — not assessed.
- **Tawi-Tawi data gap:** 100yr flood data absent from NOAA source; flood exposure in Tawi-Tawi underestimated.
- **KBA as NIPAS proxy:** WDPA dataset unavailable; KBAs do not identically map to NIPAS-designated areas.

2. REVIEW OF RELATED LITERATURE AND STUDIES

This review synthesizes four intersecting literature domains: GIS-based nuclear siting, microreactor technology for off-grid applications, Philippine off-grid electrification, and machine learning in energy site selection.

2.1 GIS-Based Nuclear Facility Siting

A bibliometric review of NPP siting research from 1968–2024 identified GIS combined with AHP as the current methodological frontier (Susiaty, et. al., 2025). The two-phase pipeline — hard

Boolean exclusion of hazard zones followed by soft weighted scoring of surviving pixels – was established for Egypt (Nematollahi et al., 2014) and has since been applied to Turkey (Kaya et al., 2025), Indonesia (Shafii, et. al, 2025), and the US (OR-SAGE) (Omitaomu, et. al., 2022). The IAEA's Safety Standards SSG-35 provides the normative exclusion criteria framework adopted in this study (IAEA, 2014).

2.2 Microreactor Technology and Off-Grid Applications

Microreactor techno-economic studies confirm cost overlap with diesel generation (USD 0.14–0.41/kWh) and establish the 20% grid penetration rule – a single generator should not exceed 20% of total grid capacity – which informs the sizing methodology in this study (Mignacca & Locatelli, 2023). A comprehensive technology review identifies the primary challenges as fuel supply immaturity, licensing frameworks, and proliferation risk (Testoni, et. al., 2021). The ASEAN nuclear momentum (Nuclear Business Platform, 2025) and Meralco-USNC cooperation agreement (World Nuclear Association, 2026) provide the commercial context for this study's Philippine application.

2.3 Philippine Off-Grid Electrification

The 634-island HRES study (Pascasio et al., 2022) provides the per-capita consumption calibration data used in the methodology and establishes that smallest islands remain financially unsustainable without subsidization – the demand stratum where microreactors are most relevant. The 2024 study of 208 island mini-grids confirms that market dynamics alone cannot achieve full renewable penetration and that alternative baseload technologies are necessary (Castro et al., 2024). The 2026 literature documents the accelerating diesel cost crisis: the Strait of Hormuz closure (Climate Smart Ventures, 2026), the 38% UC-ME rate increase (Eco-Business, 2026), and the 89.3% oil dependence of off-grid installed capacity (Rappler, 2026) collectively establish the urgency of an alternative baseload technology for Philippine island grids.

2.4 Machine Learning in Energy Site Selection

Random Forest has been validated for spatial siting classification (Breiman, 2001), with SHAP values providing interpretable per-feature contribution analysis that addresses multicollinearity artifacts in standard Gini importance scores (Lundberg & Lee, 2017; Alshari & Saleh, 2025). K-Means clustering with silhouette validation has been applied to renewable energy site typology in Australia (Zhang et al., 2024) and ocean RE in Malaysia (Idris et al., 2023); this study extends the approach by including demand_mwe as a clustering feature to produce deployment-typed rather than geography-typed archetypes.

3. METHODOLOGY

The methodology follows a five-phase pipeline: (1) hazard exclusion mapping, (2) multi-criteria suitability scoring, (3) Random Forest classification, (4) K-Means clustering, and (5) Energy Coverage Ratio analysis.

3.1 Study Area and Spatial Framework

The study area encompasses the entire Philippines – 7,641 islands, ~300,000 km² land area, bounded by 114–128°E and 4–22°N, defined by the NAMRIA Level 0 administrative boundary (OCHA/NAMRIA, 2023). All analysis was conducted at 1 km spatial resolution (working grid: 1,600 × 2,000 pixels), achieved by downsampling from the 100m Meta/CIESIN reference grid (Meta & CIESIN, 2020). We assume that all analysis at 1 km resolution due to RAM constraints on a consumer laptop (32 GB). This downsampling may lose spatial detail for very small islands and

narrow coastal strips. The 27.8% exclusion figure and all site coordinates are estimates at 1 km precision.

3.2 Data Sources

Eleven open-access GIS datasets were assembled. Table 1 summarizes all data sources.

Table 1. Open-access GIS datasets used for the study.

Layer	Source	Repository	Format	Use in study
Admin boundaries	PSA/NAMRIA 2023	HDX (cod-ab-phl)	Shapefile	Clipping, coastline, province labels
Elevation/DEM	CGIAR-CSI SRTM v4.1	HDX	Grid (90m)	Slope, terrain exclusion
Population density	Meta/CIESIN 2020	HDX	GeoTIFF ~100m	Demand estimation, reference grid
Flood hazard (100yr)	Project NOAA/UP-DREAM	Hugging Face mirror	Shapefile (80 prov.)	Flood exclusion
Landslide hazard	Project NOAA/UP-DREAM	Hugging Face mirror	Shapefile (80 prov.)	Landslide exclusion
Active faults	GEM Foundation	GitHub	GeoPackage	Seismic exclusion buffer
Volcanoes	Smithsonian GVP	volcano.si.edu	XLS	Volcanic exclusion buffer
Protected areas	KBA dataset	HDX	Shapefile	Environmental exclusion
Roads	HOT OSM Export	HDX (hotosm_phl_roads)	Shapefile	Infrastructure proximity
Ports/harbors	HOT OSM Export	HDX (hotosm_phl_sea_ports)	Shapefile	Island logistics proximity
Rivers	NAMRIA 250k	HDX	Shapefile	Freshwater/cooling proximity

3.3 Phase 1 – Hazard Exclusion Mapping

Seven binary exclusion layers were constructed following the two-phase hard/soft GIS-MCDM pipeline (Nematollahi et al., 2014) and consistent with IAEA Safety Standards SSG-35 (IAEA, 2014). Each layer was rasterized at the reference grid resolution. All layers were combined using Boolean OR logic – any single exclusion flags the pixel as excluded. Table 2 summarizes the criteria.

Table 2. GIS-based hazard exclusion for proposed sites.

Hazard	Exclusion criterion	Source	Justification
Seismic	5 km buffer around active faults	GEM, 2024	IAEA SSG-35: active fault proximity exclusion
Volcanic	10 km (eruption documented), 5 km (others)	GVP, 2026	PHIVOLCS permanent danger zone precedents; tiered by activity level
Flood	100-year return period flood zones	NOAH, 2024	IAEA SSG-35: flooding as siting constraint; 100yr is standard engineering design flood
Landslide	Landslide Hazards zones	NOAH, 2024	Ground instability incompatible with heavy reactor infrastructure
Protected areas	Key Biodiversity Areas (KBA) as NIPAS proxy	KBA/HDX, 2024	IAEA SSG-35: environmental land use constraint; NIPAS-designated areas excluded
Slope	Terrain exceeding 15°	CGIAR-CSI SRTM	Standard civil engineering threshold for heavy infrastructure construction; consistent with OR-SAGE (Omitaomu, et. al, 2022) and Indonesia siting study (Shafii, et. al, 2025)
Coastal/Tsunami	2 km inland buffer from coastline	Derived from NAMRIA adm0	Conservative tsunami inundation proxy; IAEA SSG-35 coastal siting guidance

Assumptions used in this step are: (1) The 5 km seismic buffer is a conservative fixed distance applied uniformly to all fault classifications. In practice, IAEA guidance recommends probabilistic seismic hazard analysis (PSHA) for site-specific assessment. A fixed buffer is a GIS-appropriate proxy for regional screening but may over- or under-exclude in areas with fault segments of varying slip rates. (2) The 15° slope threshold for construction infeasibility is derived from civil engineering practice for heavy infrastructure and is consistent with analogous GIS siting studies (OR-SAGE, Indonesia). It does not account for terrain leveling that may be feasible for specific microreactor designs with smaller footprints. (3) The WDPA Philippines dataset was unavailable on HDX at time of analysis; Key Biodiversity Areas (KBA) were used as a NIPAS proxy. KBAs overlap substantially with NIPAS designations but are not identical — some NIPAS-excluded areas may not be captured by KBAs and vice versa. And (4) Tawi-Tawi province 100-year flood data was absent from the NOAH Hugging Face mirror dataset. Flood exclusion in Tawi-Tawi is therefore underestimated.

3.4 Phase 2 – Multi-Criteria Suitability Scoring

Following the hard exclusion phase, surviving viable pixels were scored, and then normalized into a 0-100 scale using four criteria. The two-phase structure follows established GIS-MCDM nuclear

siting precedent (Nematollahi et al., 2014; Omitaomu, et. al, 2022). Exponential decay functions were used rather than the stepped brackets common in earlier studies, because stepped brackets create artificial discontinuities: a site at 4.9 km from a population center would score identically to one at 0.1 km, then drop sharply at 5 km. Exponential decay produces continuously differentiated scores that are physically more realistic – proximity benefit decays gradually rather than discontinuously.

Criterion	Weight	Scoring function	Justification
Demand proximity	30 pts	$30 \times \exp(-\text{dist_pop} / 15 \text{ km})$	Primary driver of microreactor economic viability; communities with existing demand justify deployment cost (Mignacca & Locatelli, 2023)
Port logistics access	20 pts	$20 \times \exp(-\text{dist_port} / 30 \text{ km})$	Ports are the only practical delivery mode for factory-assembled reactor modules and fuel in archipelagic settings; longer half-distance reflects that port access within 30 km is still logistically viable
Road access	10 pts	$10 \times \exp(-\text{dist_road} / 10 \text{ km})$	Construction, maintenance, and security logistics; even off-grid deployment requires road access
Freshwater / cooling water	10 pts	$10 \times \exp(-\text{dist_river} / 5 \text{ km})$	IAEA SSG-35 requires cooling water availability; tighter half-distance reflects that freshwater must be reliably close
Slope penalty	-0 to -10	$-10 \times \text{clip}(\text{slope}/15^\circ, 0, 1)$	Continuous terrain penalty before the exclusion threshold; rewards very flat land

For this step, we assume that the half-score distances (15 km for population, 30 km for ports, 10 km for roads, 5 km for rivers) are author-defined parameters calibrated from the following rationale: the 15 km population distance is consistent with the 15 km service radius used in the ECR demand estimation (Pascasio et al., 2022); the 30 km port distance reflects practical barge logistics feasibility in Philippine archipelagic conditions (no direct literature citation exists for this specific value). A sensitivity analysis of these parameters would strengthen the findings but was beyond the project scope.

3.5 Phase 3 – Random Forest Classification

A LazyPredict comparison of 15 classifiers was conducted prior to model selection (LazyPredict v0.2.7). Results showed ExtraTrees at 88.3%, Random Forest at 87.8%, and BaggingClassifier at 86.1%. Random Forest was selected over ExtraTrees despite its marginally lower accuracy because: (1) RF is better established in the siting literature (Breiman, 2001); (2) RF with SHAP produces more interpretable feature importance than ExtraTrees; and (3) the 0.5% accuracy

difference is within cross-validation noise (CV std: 1.8%). We assume that class labels (Unsuitable/Marginal/Suitable) are derived from the MCDM suitability scores, which are themselves constructed from the author-defined scoring criteria and weights. This means the RF classifier is trained to learn the spatial pattern of the MCDM score thresholds, not independently derived 'ground truth' labels. The RF adds value by identifying which features most strongly predict those patterns across unseen pixels – but it does not independently validate the scoring framework.

The features chosen and the model configurations are as follows:

- 7 features: dist_pop, dist_port, dist_road, dist_river, pop_density, slope, elevation
- n_estimators=200, max_depth=15, random_state=42; 80/20 stratified train-test split; 5-fold cross-validation
- SHAP TreeExplainer for feature importance (Lundberg & Lee, 2017) – chosen over Gini importance to handle multicollinearity artifacts

3.6 Phase 4 – K-Means Clustering

K-Means clustering was applied to the top 200 candidate sites to identify deployment archetypes (Zhang et al., 2024). A key methodological distinction from prior energy site clustering studies is the inclusion of demand_mwe as a clustering feature alongside infrastructure variables. This ensures that archetypes are typed by both infrastructure context and demand level – producing deployment-informed clusters rather than purely geographic groupings.

Optimal k was determined using elbow method and silhouette analysis (Rousseeuw, 1987). All features were standardized using StandardScaler before clustering.

3.7 Phase 5 – Energy Demand Analysis and ECR

We estimate the energy demand as follows:

- **Population served:** Meta/CIESIN population density (people/km²) × service area ($\pi \times 15^2$ km = 707 km²). (Meta & CIESIN, 2020)
- **Per-capita consumption calibration:** from Pascasio et al. (2022) – 500 kWh/yr (remote, dist_pop > 20 km), 800 kWh/yr (coastal), 1,200 kWh/yr (peri-urban, dist_pop < 5 km)
- **Annual demand:** demand_kwh = pop_served × kWh/capita/yr
- **Demand in MWe:** demand_mwe = demand_kwh ÷ 8,760 hours ÷ 1,000

Per-capita consumption values (500 / 800 / 1,200 kWh/yr) are calibrated from island-level aggregates in Pascasio et al. (2022), not from metered consumption surveys of individual communities. Actual consumption may differ, particularly for communities with economic activities not reflected in residential consumption proxies. The 15 km service radius assumes that a microreactor can supply electricity to all communities within 15 km via distribution infrastructure. This assumes either existing distribution lines or new lines are buildable – which may not hold for all island geographies. The radius is a policy modeling assumption, not a technical guarantee. Additionally, population density from Meta/CIESIN 2020 data does not account for population growth between 2020 and the study date (2026). Philippine off-grid island populations may have grown or declined since 2020.

Reactor sizes are limited to the discrete commercial unit sizes assumed available (1, 2, 5, 10, 20 MWe). In practice, specific microreactor designs have fixed output ratings that may not match these rounded sizes. The 1 MWe minimum unit is based on current early commercial designs; smaller designs (0.1–0.5 MWe) exist in development but were excluded for conservatism.

As for the reactor sizing, this pipeline was followed:

- **Reactor capacity:** $\text{demand_mwe} \times 1.2$ (20% growth buffer), rounded to nearest commercial unit: 1, 2, 5, 10, or 20 MWe
- **Basis for 20% buffer:** derived from the grid penetration rule in Mignacca & Locatelli (2023) – no single generator should exceed 20% of total grid capacity
- **ECR:** $\text{ECR} = \text{reactor_capacity} \div \text{demand_mwe}$
- **ECR categories:** Balanced (1.0–1.5) · Adequate (0.7–1.0) · Oversupply (>1.5) · Undersupply (<0.7)
- **Final ranking:** 60% normalized suitability score + 40% ECR match score (proximity to ideal ECR of 1.2)

4. RESULTS AND DISCUSSION

4.1 Hazard Exclusion Results

The master exclusion mask excluded 27.8% of Philippine land area, leaving 72.2% viable. Table 3 shows the breakdown by layer.

Table 3. GIS exclusion of Philippine sites for microreactor siting.

Exclusion layer	% Excluded	Source	Geographic pattern
Protected areas (KBA proxy)	16.4%	KBA/HDX 2024	Sierra Madre, Palawan interior, Mindanao highlands
Landslide (NOAH)	6.6%	UP-DREAM 2024	Mountainous interior Luzon and Mindanao
Seismic – GEM faults	5.1%	GEM Foundation 2024	Philippine Fault Zone corridor (Luzon–Leyte)
Coastal buffer (2 km)	4.1%	Derived NAMRIA adm0	Coastlines of all island groups
Slope >15°	2.6%	CGIAR-CSI SRTM	Cordillera, Mt. Apo, Leyte highlands
Flood 100yr (NOAH)	2.2%	UP-DREAM 2024	Central Luzon plain, major river deltas
Volcanic (23 vents)	0.3%	Smithsonian GVP 2026	Taal, Mayon, Bulusan, Kanlaon, others
TOTAL EXCLUDED	27.8%	Boolean OR	Viable land: 72.2%

The dominance of protected areas (16.4%) reflects the Philippines' extensive NIPAS designation of biodiversity-critical zones. The seismic exclusion (5.1%) follows the Philippine Fault Zone corridor documented by GEM (GEM Foundation, 2024). The 27.8% total exclusion rate is higher than the 1.78% found for Bangka Belitung (Shafii, et. al, 2025), reflecting more comprehensive exclusion criteria (7 layers including protected areas, which the Indonesia study did not apply at the same scale) and the Philippines' extensive NIPAS coverage.

4.2 Suitability Score Distribution

At 1 km resolution, suitability scores ranged 0–100 (mean 18.8, median 13.4) across viable pixels – strongly right-skewed, reflecting that most inland terrain scores low on demand proximity and port access. High-scoring sites (≥ 70) account for 2.19% of viable pixels, concentrated exclusively within 15 km of coastlines. After spatial thinning (best pixel per 0.5° grid cell), 200 candidate sites were identified. Figure 1 shows the areas in the Philippine archipelago suitable for microreactors based on the exclusion criteria and the MCDM scoring. The stars on the figure are the top 20 sites where it is possible to erect them.

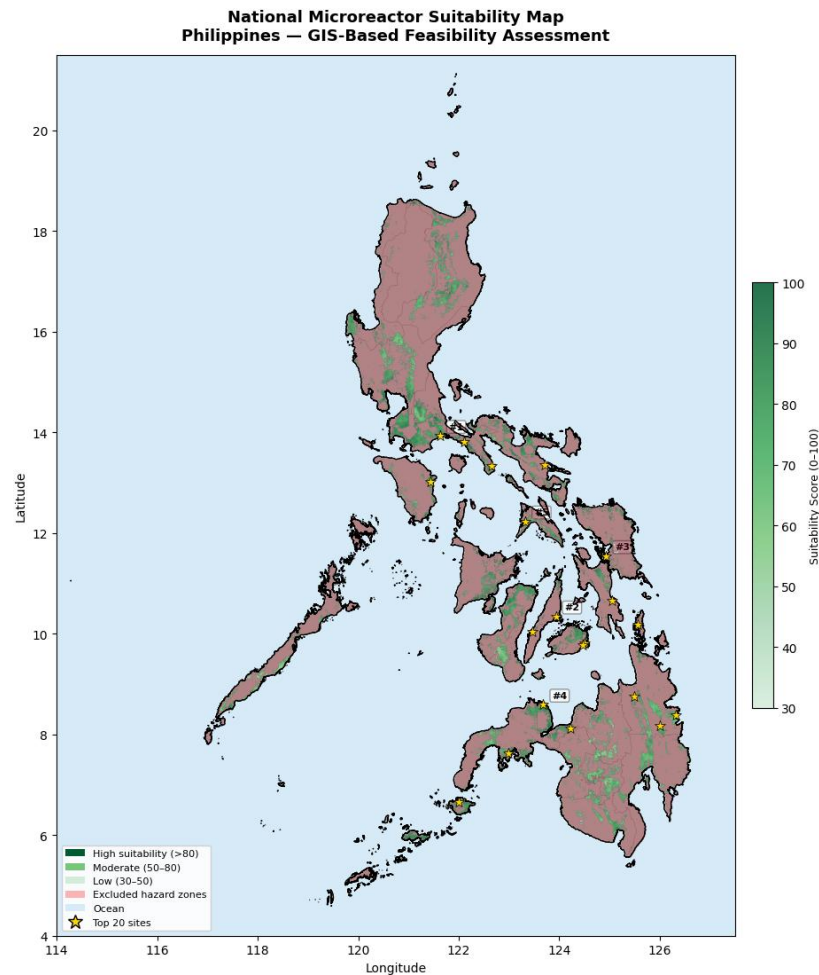


Figure 1. Microreactor suitability map based on GIS exclusion and MCDM scoring

4.3 Random Forest Classification Results

The following section presents the results of the model selection process. The selected model is then evaluated for a five-fold cross-validation.

Table 4. LazyPredict model selection results.

Classifier	Accuracy	Balanced Acc.	ROC AUC	F1 Score
ExtraTreesClassifier	0.883	0.883	0.935	0.878

RandomForestClassifier ✓	0.878	0.878	0.937	0.872
BaggingClassifier	0.861	0.861	0.921	0.858
GaussianNB	0.828	0.828	0.908	0.824
AdaBoostClassifier	0.822	0.822	0.849	0.814

Table 5. Random Forest Classifier results.

Class	Precision	Recall	F1-Score	Support
Class 0 – Unsuitable	0.93	0.68	0.79	60
Class 1 – Marginal	0.88	0.95	0.91	60
Class 2 – Suitable	0.85	1.00	0.92	60
Overall / 5-fold CV	0.88	87.8% / 89.0%±1.8%	0.87	180

The 100% recall for Class 2 (Suitable) is the most important result for a siting screening application: the model never misses a genuinely suitable location. The 68% recall for Class 0 (Unsuitable) reflects conservative classification – the model errs toward inclusion, which is appropriate for preliminary screening where false negatives (missed suitable sites) are more costly than false positives (Breiman, 2001). Note that this model is created as a tool for microreactor siting decision makers, to determine if a proposed site is suitable for the microreactor. Figure 2 shows a side-by-side comparison of the actual siting map vs the map generated from the random forest classifier alone.

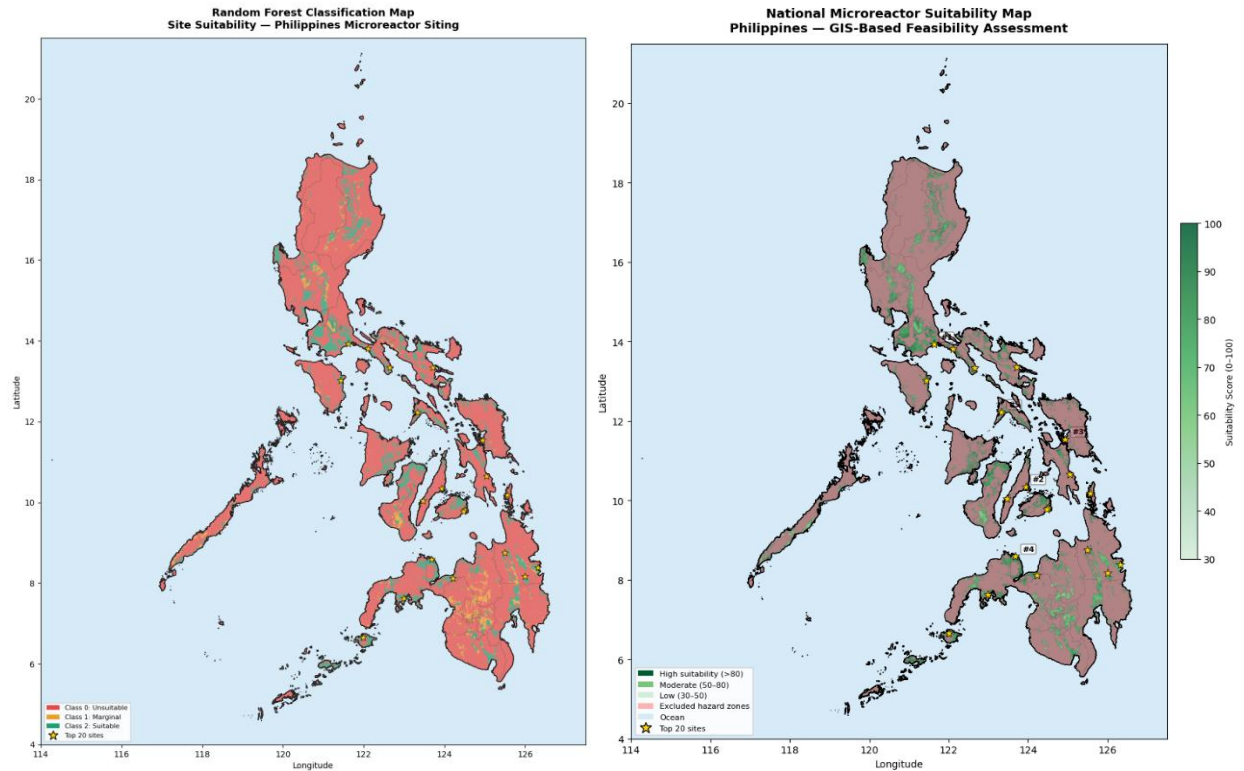


Figure 2. Side-by-side comparison of the GIS-based site mapping vs the Random Forest classifier based site mapping.

4.4 K-Means Cluster Archetypes

K-Means clustering with $k=3$ (selected over statistically optimal $k=2$ for deployment interpretability) identified three distinct deployment archetypes. These archetypes emerged from the data — they were not defined a priori. The archetype names are interpretations of the cluster centroid profiles, supported by the mean feature values in Table 6.

Optimal k was determined using elbow method and silhouette analysis (Rousseeuw, 1987). As shown in Figure 3, while $k=2$ produced the highest silhouette score, $k=3$ was selected because: (1) the 180/20 split at $k=2$ is too lopsided to support meaningful policy differentiation; (2) $k=3$ produces three distinct deployment tiers that map to three different reactor size strategies; and (3) the silhouette score difference between $k=2$ and $k=3$ (0.519 vs 0.408) represents a moderate reduction that is acceptable given the interpretability gain. All features were standardized using StandardScaler before clustering. Figure 4 shows the top 200 identified sites, and their clusters, accordingly.

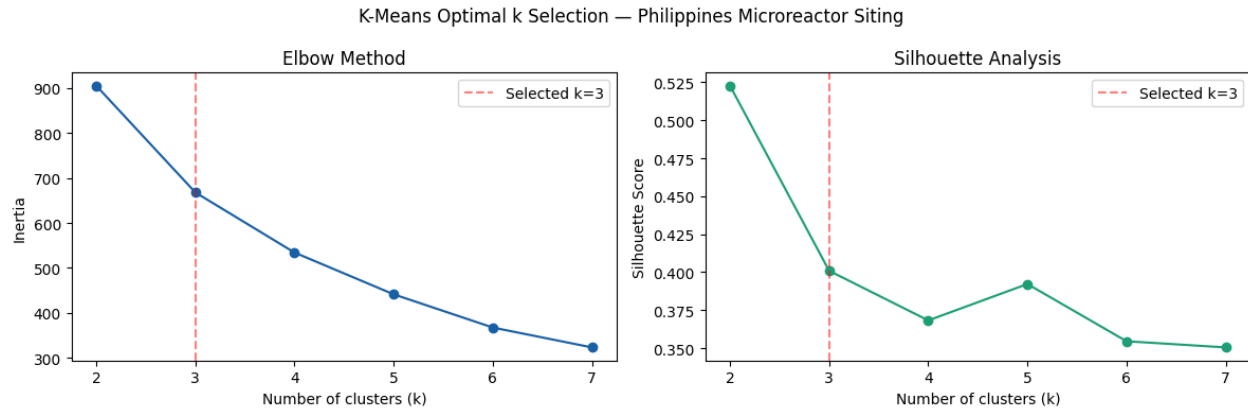


Figure 3. Elbow method and silhouette analysis for optimal number of clusters.

Table 6. Summary of formed clusters within the top 200 sites using k-means.

Cluster	Archetype	n (share)	dist_pop	dist_port	Demand	Reactor
0	Coastal Island Hub	144 (72%)	2.6 km	17.5 km	0.70 MWe	1–2 MWe
1	Small Coastal Community	20 (10%)	7.7 km	7.1 km	0.13 MWe	1 MWe
2	Peri-Urban Grid Anchor	36 (18%)	2.8 km	11.1 km	2.98 MWe	2–10 MWe

Cluster 0 – Coastal Island Hub (144 sites, 72%)

These are the most common site type in the Philippines off-grid landscape. Characterized by very close proximity to population centers (mean 2.6 km), moderate port access (mean 17.5 km), and low to moderate electricity demand (mean 0.70 MWe), these sites represent accessible coastal communities with established road infrastructure but no major commercial port nearby. They are typically provincial coastal towns or smaller island municipalities that already have municipal-level electricity demand and existing distribution infrastructure. The 17.5 km port distance means barge access is feasible but not immediate – reactor module delivery is achievable but requires advance logistics planning. Standard 1–2 MWe microreactor units are appropriate, typically resulting in Balanced ECR (1.0–1.5). These sites represent the largest and most tractable deployment opportunity – the highest density of the Philippine SIIG electrification challenge.

Cluster 1 – Small Coastal Community (20 sites, 10%)

These sites are the most geographically isolated of the three archetypes. They have the largest distance to population centers (mean 7.7 km) and the lowest electricity demand (mean 0.13 MWe) – indicating small, somewhat dispersed communities. Paradoxically, they have the closest port access of the three clusters (mean 7.1 km), which suggests these communities are maritime-dependent settlements – fishing villages, small island barangays, or communities that rely on regular boat connections rather than road access. The very low demand (0.13 MWe) means that even the smallest available 1 MWe microreactor unit would result in substantial oversupply (ECR of 5–10). These sites are not economically viable as standalone deployments in conventional terms – they require a multi-island cable networking strategy where surplus power is distributed

to neighboring communities, or they should be evaluated for pairing with a smaller mobile nuclear battery design if one becomes commercially available. They represent a longer-term Phase 3 deployment target.

Cluster 2 – Peri-Urban Grid Anchor (36 sites, 18%)

These sites have the highest average suitability scores (91.4) and the highest electricity demand (mean 2.98 MWe) of the three archetypes. They are characterized by very close proximity to population centers (mean 2.8 km) and moderate port access (mean 11.1 km). These are near-city coastal communities – located adjacent to or within the catchment areas of provincial capitals, regional centers, or rapidly growing secondary cities. The higher demand reflects denser population and greater commercial and industrial activity. These sites are candidates for larger 2–10 MWe microreactor units, depending on specific local demand calculations, and are best suited for integration with regional microgrids rather than standalone off-grid deployment. They represent the strongest economic case – high demand means better cost-per-kWh and the strongest justification for capital investment. These sites should be prioritized for Phase 2 deployment after the Coastal Island Hubs establish operational precedent.

All the top 200 identified sites are mapped according to their respective clusters for visualization purposes in Figure 4.

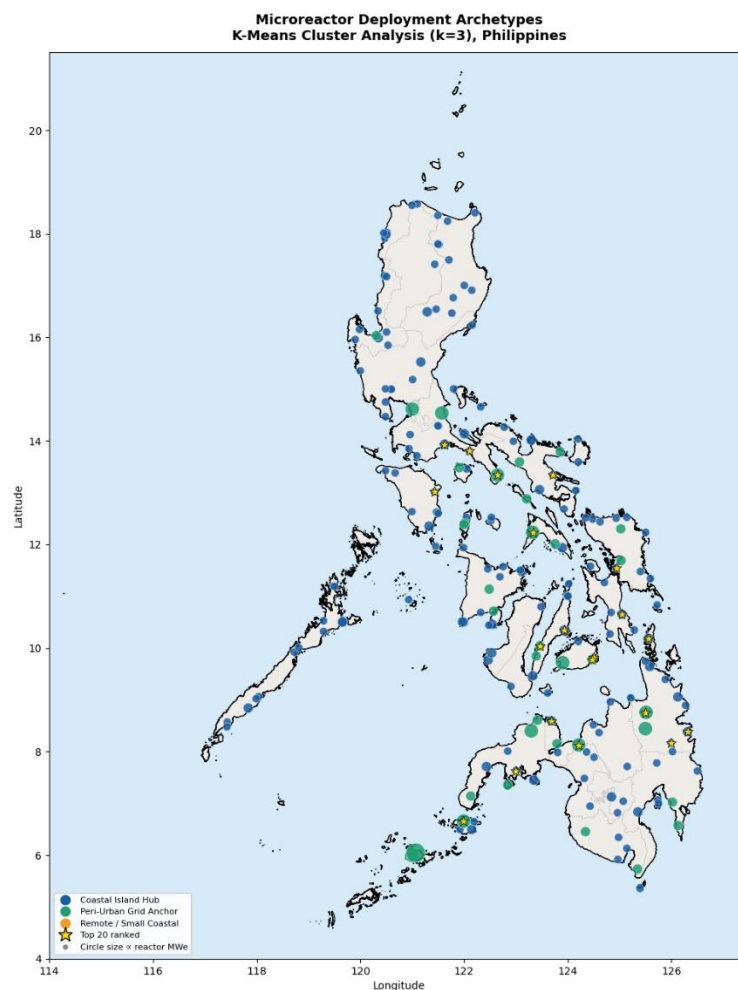


Figure 4. Top 200 sites according to their respective clusters.

4.5 Energy Coverage Ratio Distribution

ECR values ranged from 0.71 to 14.53 (mean 2.23) across 200 sites. The high mean reflects the minimum commercial unit constraint: even the smallest available reactor (1 MWe) exceeds demand for the smallest communities. Figure 5 shows the map of the top 200 sites and the ECR category of the proposed reactors for each, while Table 7 summarizes the distribution of the ECRs throughout the sites.

Table 7. Energy coverage ratio (ECR) distribution for the top 200 microreactor viable sites.

ECR Category	Range	Sites (n / %)	Implication
Balanced	1.0 – 1.5	60 (30%)	Well-matched reactor capacity to local demand; direct deployment feasible
Oversupply	> 1.5	89 (44.5%)	Minimum 1 MWe unit exceeds demand; consider multi-island cable network to export surplus
Adequate	0.7 – 1.0	30 (15%)	Demand met with limited growth margin; feasible deployment
Undersupply	< 0.7	21 (10.5%)	Reactor undersizes demand; dual units or next size up needed

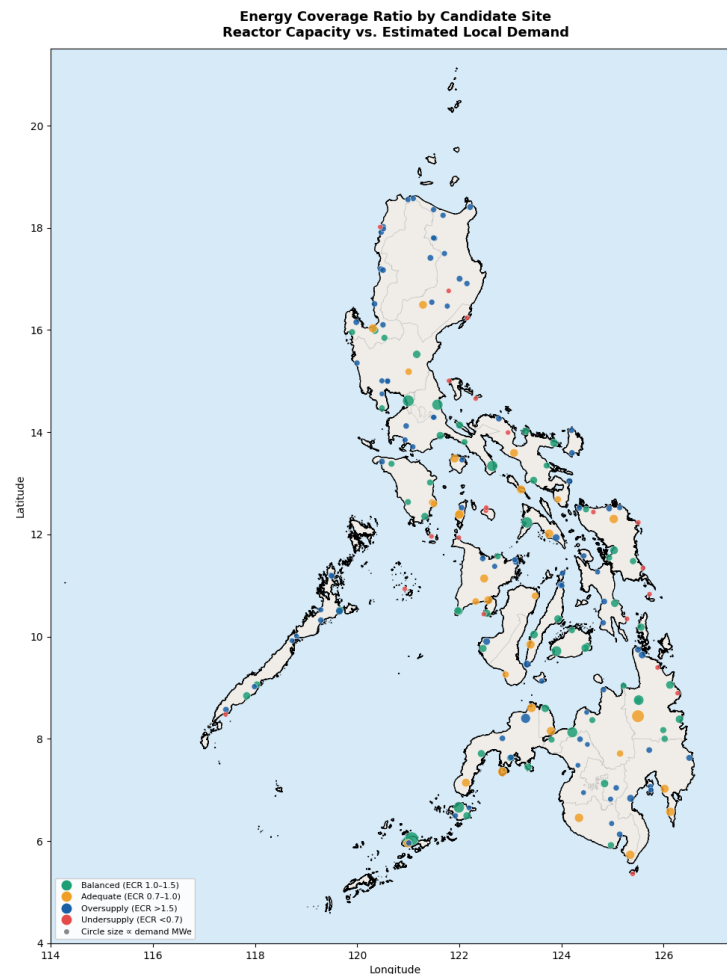


Figure 5. Top 200 sites by Energy Coverage Ratio.

All top 20 ranked sites achieved Balanced ECR – an emergent result confirming that the composite ranking (60% suitability + 40% ECR match) effectively filters for deployable sites rather than merely technically accessible ones.

4.6 Top 20 Ranked Sites

The top 20 sites span all three major island groups. Table 8 reports the top 10, and Figure 6 shows where in the Philippine map they are located.

The reactor distribution across top 200 sites are: 1 MWe (134, 67%) · 2 MWe (54, 27%) · 5 MWe (11, 5.5%) · 10 MWe (1, 0.5%). The Philippine off-grid demand profile is dominated by sites requiring 1–2 MWe – matching standard small-unit microreactor designs.

Table 8. Top 20 most viable microreactor sites.

Rank	Province	Lat	Lon	Score	Archetype	MWe	ECR	Region
1	Quezon	13.93°N	121.63°E	99.2	Coastal Hub	2	1.24	IV-A (Calabarzon)
2	Cebu	10.35°N	123.93°E	97.7	Coastal Hub	2	1.24	VII (Central Visayas)
3	Samar (W.)	11.54°N	124.93°E	96.9	Coastal Hub	1	1.23	VIII (Eastern Visayas)
4	Misamis Occ.	8.59°N	123.68°E	96.9	Coastal Hub	2	1.15	X (N. Mindanao)
5	Masbate	12.23°N	123.33°E	96.0	Peri-Urban	5	1.14	V (Bicol)
6	Leyte	10.65°N	125.04°E	94.6	Coastal Hub	2	1.16	VIII (Eastern Visayas)
7	City of Isabela	6.66°N	121.99°E	94.3	Peri-Urban	5	1.16	IX (Zamboanga)
8	Quezon	13.34°N	122.64°E	91.8	Peri-Urban	5	1.22	IV-A (Calabarzon)
9	Zamb. Sibugay	7.63°N	122.99°E	92.9	Coastal Hub	1	1.16	IX (Zamboanga)
10	Cebu	10.04°N	123.46°E	96.6	Peri-Urban	2	1.07	VII (Central Visayas)

Top 20 Ranked Microreactor Sites Philippines — Deployment Priorities

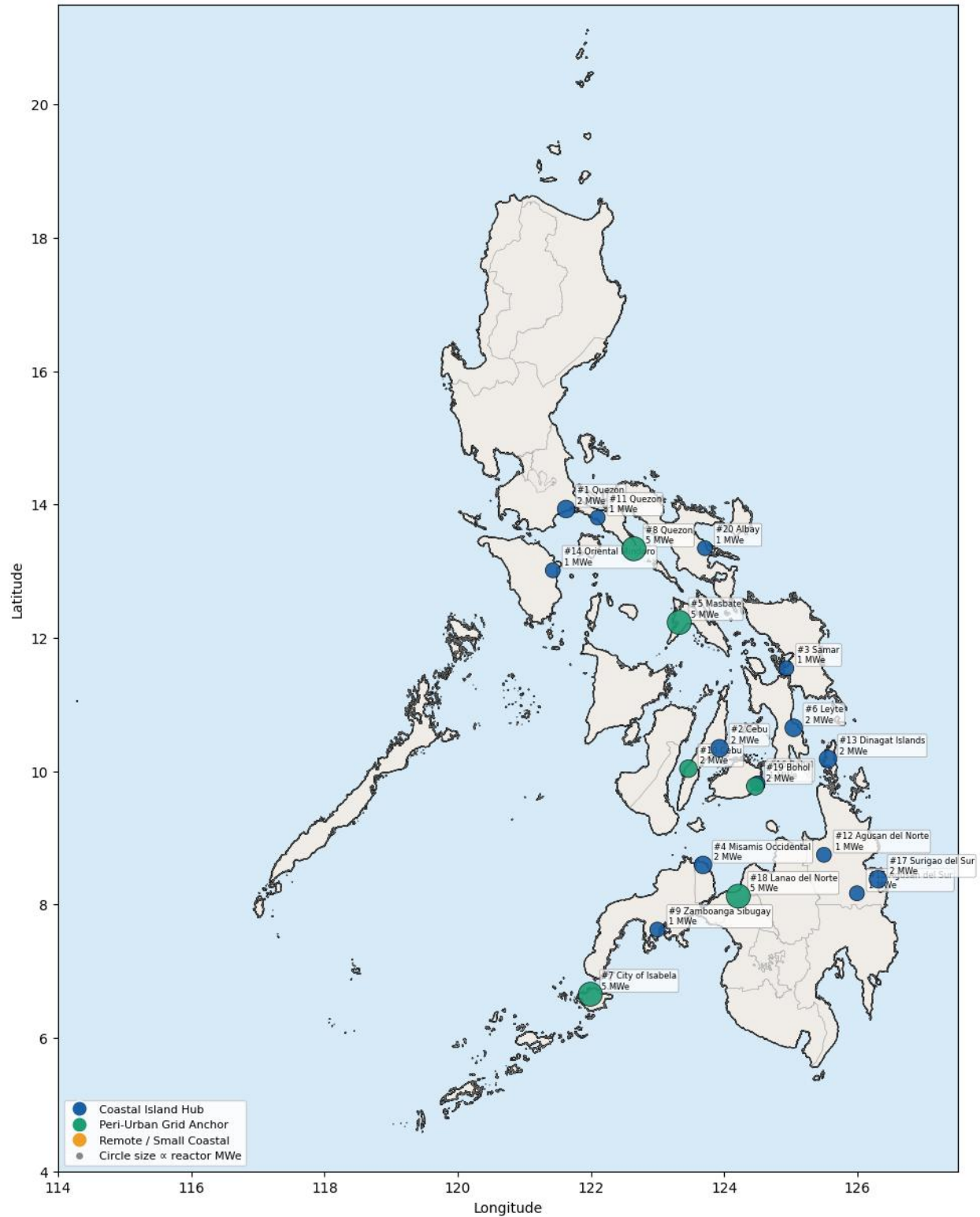


Figure 6. Top 20 most viable microreactor sites in the Philippines.

4.7 What Makes a Good Microreactor Site — SHAP Interpretation

SHAP analysis (Lundberg & Lee, 2017) provides an empirically grounded answer to what constitutes a suitable Philippine microreactor site — one not dependent on assumed weights but derived from the spatial data structure. The four dominant SHAP drivers are directly actionable for deployment planning.

Demand proximity (SHAP 0.1513): Sites close to communities score highest across all scoring dimensions. This confirms that the primary deployment justification is replacing existing diesel demand — not creating new supply for undeveloped areas. The practical implication is that the DOE should prioritize communities with documented NPC-SPUG meter connections and existing consumption records, not theoretical future demand.

Road access (SHAP 0.1232): Infrastructure connectivity is nearly as important as demand proximity — a finding that challenges the assumption that off-grid deployment is insensitive to road quality. Even a factory-assembled microreactor requires road access for the final delivery leg, site preparation, periodic maintenance, waste management, and security. Communities accessible only by boat face substantially higher deployment costs not captured in the reactor unit economics.

Freshwater availability (SHAP 0.0829): River proximity ranks third, consistent with IAEA SSG-35 requirements for cooling water at all reactor types. Even passive-safety microreactor designs that use air-cooled heat rejection may require a nearby water source for emergency cooling, secondary systems, or on-site operations.

Port access (SHAP 0.0678): Port access is statistically significant at rank 4 despite being absent from all continental nuclear siting frameworks reviewed. This validates the core archipelagic novelty claim of this study: in an island setting, ports are the logistics backbone for reactor module delivery — there is no other practical option for transporting a factory-assembled reactor unit to a remote island. The finding has direct regulatory implications: site-specific EIA for Philippine microreactors should include a port logistics assessment as a standard siting criterion.

4.8 Reactor Sizing and ECR Findings

The results highly suggest that 1–2 MWe is the appropriate starting unit. 134 of 200 candidate sites (67%) require only 1 MWe. Philippine off-grid islands are characterized by small communities with peak demand well below 1.5 MWe. Oversizing with larger units wastes capital and violates the 20% grid penetration rule for standalone deployment (Mignacca & Locatelli, 2023).

In fact, 44.5% of sites are oversupplied at 1 MWe. The minimum commercial unit exceeds demand for nearly half the sites. These sites require a multi-island cable networking strategy. A 1 MWe microreactor in Quezon with ECR 1.24 generates approximately 0.2 MWe of exportable surplus — enough to supply 3–4 neighboring barangays via submarine cable at minimal incremental cost.

Only 1 site in 200 requires 10 MWe size microreactor. Large microreactor units are not appropriate for the Philippine off-grid context. The demand landscape is fragmented across many small communities, not concentrated at a few large loads.

4.9 Policy Implications — Philippine Nuclear Implementation

- **The regulatory bottleneck is resolving.** The establishment of PhilATOM under RA 12305 (September 2025) (PhilATOM, 2025) and the DOE's plan to begin accepting nuclear license applications in 2026 (Pinsent Masons, 2026) remove the primary institutional barrier that previously made Philippine microreactor deployment impossible. The USTDA

commitment of USD 2.7 million for Meralco PowerGen's SMR feasibility study (World Nuclear News, 2026b) and Meralco's existing USNC cooperation agreement (World Nuclear Association, 2026) confirm that commercial interest exists. This study's GIS evidence base is timely: the DOE has stated it will present the licensing flowchart to prospective nuclear proponents (World Nuclear News, 2026a) a site-selection framework like this one is precisely the technical evidence such proponents need.

- **The UC-ME framework provides an institutional pathway.** The top 20 ranked sites are concentrated in exactly the provinces served by NPC-SPUG's missionary electrification mandate. Transitioning UC-ME funding from diesel subsidy to capital financing of microreactor deployments would not require new legislation — only policy re-direction. The 38% UC-ME rate increase in February 2026 (Eco-Business, 2026) and the PHP 2.5 billion emergency budget allocation for NPC fuel subsidies (Eco-Business, 2026) demonstrate that the current diesel model is fiscally unsustainable and politically costly. Microreactors offer a permanent structural solution.
- **The geopolitical context strengthens the case.** The March 2026 Strait of Hormuz closure and associated diesel price surge (Climate Smart Ventures, 2026; Rappler, 2026) demonstrated in real time that 89.3% of off-grid installed capacity being oil-dependent (Rappler, 2026) represents a national energy security vulnerability. Microreactors — with 8–20-year domestic fuel cycles — would eliminate this exposure for the communities in this study's top 20 ranked sites.

5. CONCLUSION

This study addressed two research questions: where are safe and suitable microreactor sites in the Philippines, and can these sites meet local electricity demand? Both are answered affirmatively and with quantitative specificity, at a moment when the policy and regulatory context makes these answers actionable rather than merely academic.

The GIS hazard exclusion framework identified 200 candidate sites nationwide after excluding 27.8% of land area. High-scoring sites concentrate exclusively within 15 km of coastlines — a finding explained by the SHAP-validated primacy of demand proximity, road access, freshwater availability, and port logistics as siting drivers. The statistically significant port access finding (SHAP 0.0678) is the study's most distinctive contribution: it empirically confirms that island logistics feasibility is a genuine siting driver in archipelagic contexts and should be included as a standard criterion in Philippine nuclear siting guidelines.

All top 20 ranked sites achieved a Balanced ECR (1.0–1.5). Standard 1–2 MWe microreactor units are appropriate for 90% of candidate sites. The Philippine off-grid demand profile — fragmented across many small communities rather than concentrated at large loads — is precisely matched by the 1–5 MWe microreactor tier. Oversupplied sites (ECR >1.5) are candidates for multi-island submarine cable networking rather than larger reactor units.

This study represents the first machine learning-augmented microreactor siting framework for a Philippine archipelagic context and provides the GIS evidence base aligned with the DOE's 2026 nuclear licensing roadmap (Pinsent Masons, 2026), NPC-SPUG's missionary electrification mandate, and the active commercial interest of Meralco-USNC (World Nuclear Association, 2026). The deteriorating diesel cost crisis of 2026 — the 38% UC-ME rate increase (Eco-Business, 2026), the Strait of Hormuz supply shock (Climate Smart Ventures, 2026), and the 89.3% oil dependence of off-grid installed capacity (Rappler, 2026) — make the transition from diesel to microreactor baseload not merely desirable but urgent. Future work should address economic

feasibility (LCOE by site), climate resilience modeling for 2050 sea-level scenarios, and social acceptance surveys in priority communities – particularly in BARMM and historically underserved regions among the top-ranked sites.

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